Hallucination Detection System

Evaluation Report

Based on 100 Questions · Models: GLM, GPT, LLAMA, Mistral

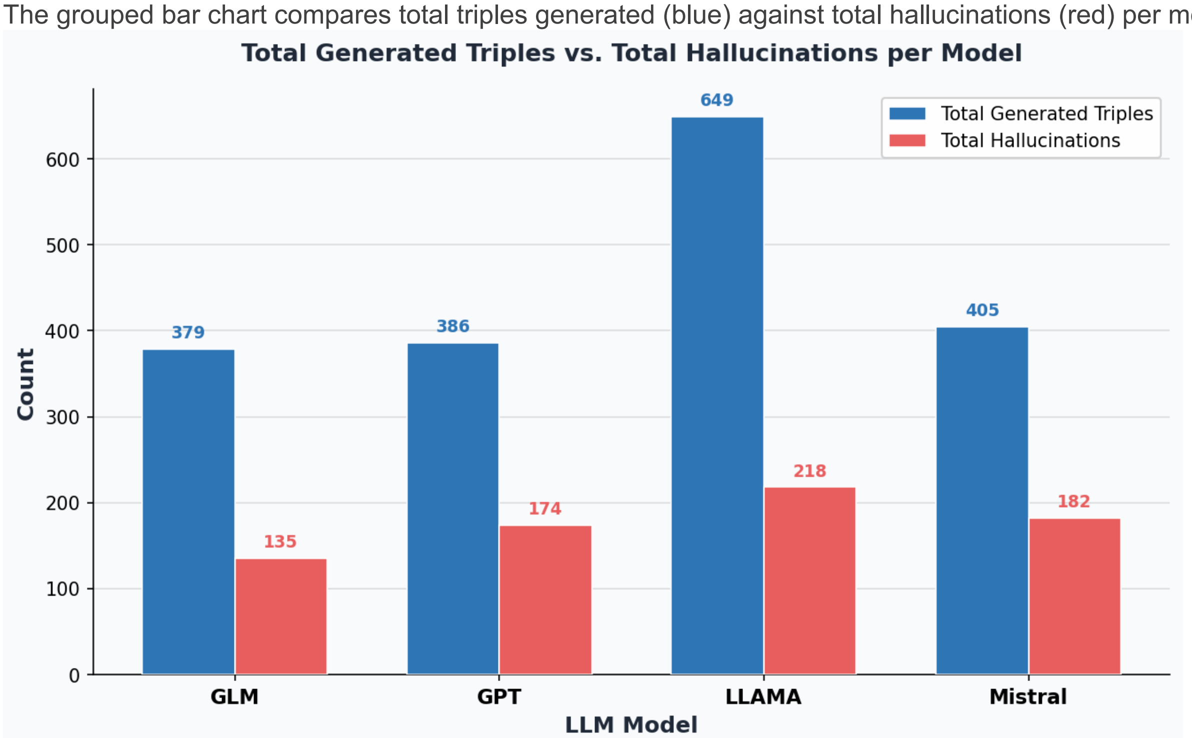
1. Hallucination Overview per Model

The table below shows the total triples generated, total hallucinations, and hallucination rate for each model.

Model	Total Generated Triples	Total Hallucinations	Hallucination Rate
GLM	379	**135**	35.62%
GPT	386	**174**	45.08%
LLAMA	649	**218**	33.59%
Mistral	405	**182**	44.94%
TOTAL	**1819**	**709**	**38.98%**

2. Generated Triples vs. Hallucinations — Visual Comparison

The grouped bar chart compares total triples generated (blue) against total hallucinations (red) per model.



3. System Accuracy — F1 Score Analysis

System accuracy is evaluated using the F1 score — the harmonic mean of Precision and Recall. TP = min(detected, actual), FP = max(0, detected − actual), FN = max(0, actual − detected).
Precision = TP / (TP + FP) Recall = TP / (TP + FN) F1 = 2 × (P × R) / (P + R)

Table 3a — Confusion Metrics per Model

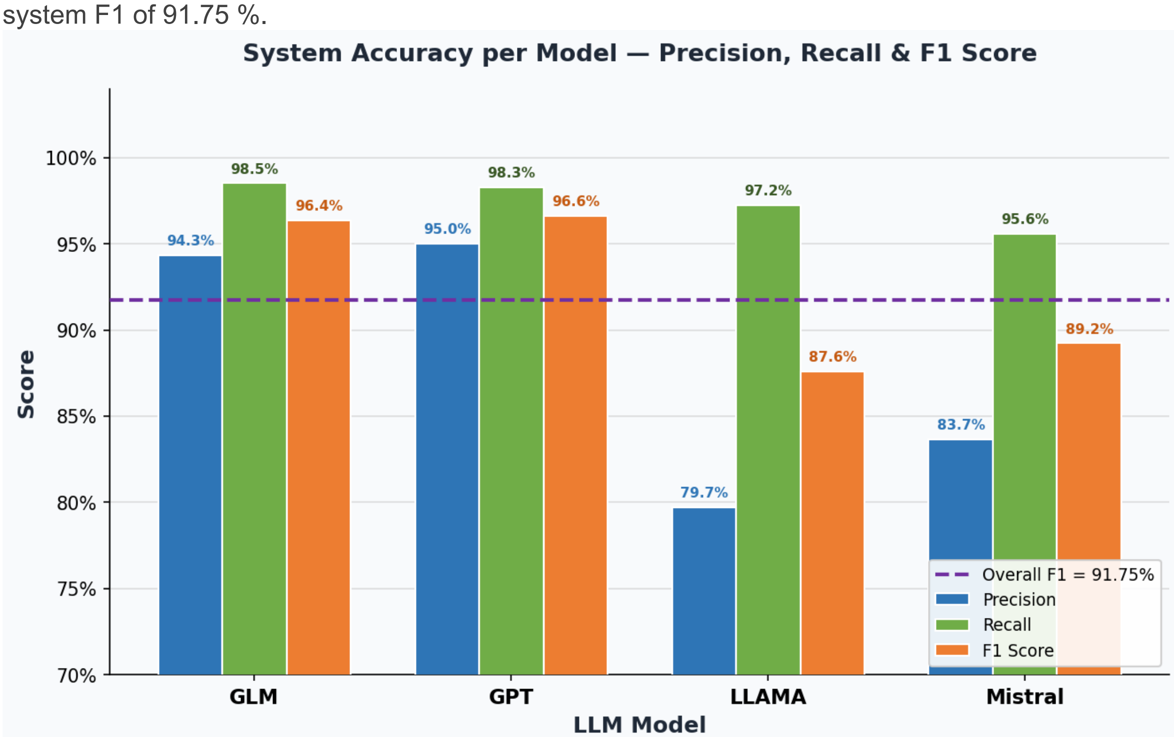
Model	TP	FP	FN	Precision	Recall	F1 Score
GLM	**133**	8	2	94.33%	98.52%	**96.38%**
GPT	**171**	9	3	95.00%	98.28%	**96.61%**
LLAMA	**212**	54	6	79.70%	97.25%	**87.60%**
Mistral	**174**	34	8	83.65%	95.60%	**89.23%**
OVERALL	**690**	**105**	**19**	**86.79%**	**97.32%**	**91.75%**

Table 3b — F1 Score Summary

Model	Precision	Recall	F1 Score
GLM	94.33%	98.52%	**96.38%**
GPT	95.00%	98.28%	**96.61%**
LLAMA	79.70%	97.25%	**87.60%**
Mistral	83.65%	95.60%	**89.23%**
SYSTEM TOTAL	**86.79%**	**97.32%**	**91.75%**

4. System Accuracy — Visual Comparison

Precision (blue), Recall (green), and F1 Score (orange) per model. The dashed purple line marks the overall system F1 of 91.75 %.



5. LLM alone Compare with LLM+System

Model	Hallucination without system	Caught by system TP	Missed FN	Detection rate
GLM	**135**	133	2	**98.52%**
GPT	**174**	171	3	**98.28%**
LLAMA	**218**	212	6	**97.25%**
Mistral	**182**	174	8	**95.60%**
OVERALL	**709**	**690**	**19**	**97.32%**

6. Types of hallucinations detected

Mistral

Hallucination Type	Generated	Detected Correctly	Not detected	Detected wrongly
Fabricated Fact	**142**	136	6	6
Rule violation	**9**	9	0	0
Rule violation from constraint	**12**	10	2	9
Predicate mismatch	**12**	12	0	0
Logical inconsistency	**7**	7	0	3
OVERALL	**182**	**174**	**8**	**32**

GPT

Hallucination Type	Generated	Detected Correctly	Not detected	Detected wrongly
Fabricated Fact	136	133	3	3
Rule violation	13	13	0	3
Rule violation from constraint	14	14	0	4
Predicate mismatch	12	12	0	0
Logical inconsistency	3	3	0	2
OVERALL	**174**	**171**	**3**	**9**

LLAMA

Hallucination Type	Generated	Detected Correctly	Not detected	Detected wrongly
Fabricated Fact	156	151	5	10
Rule violation	33	33	0	19
Rule violation from constraint	14	14	0	20
Predicate mismatch	4	3	1	4
Logical inconsistency	11	11	0	1
OVERALL	**218**	**212**	**6**	**54**

GLM

Hallucination Type	Generated	Detected Correctly	Not detected	Detected wrongly
Fabricated Fact	126	124	2	2
Rule violation	5	5	0	3
Rule violation from constraint	1	1	0	5
Predicate mismatch	0	0	0	0
Logical inconsistency	3	3	0	0
OVERALL	135	133	2	8

7. Key Insights

. The system is evaluated across 100 questions. 40 questions was focusing on questions involving single relationship from relationships in context , and 60 questions that cover 2 relationships with all possible combinations can be done from context.

. LLAMA is the Lowest hallucination rate with having largest number of hallucinations as llama tend in it’s answers to add unverifiable info or additional info not asked for that is correct.

. The types of hallucinations generated is something that cannot be controlled from the LLM output so we did not have control of number of hallucination generated from each type but we tried to stress different types of hallucinations according to it’s weight in the context , predicate mismatch is the lowest as there is only one relationship in the context that is presented for some and some not.

. The accuracy of the system is relatively high as the system is asking about closed domain and asking the LLMs for structured output so the process is easier, but with LLAMA the scenario is different as it sometimes provide additional info unstructured that cannot be detected and treated as unverifiable so it’s F1 score is the lowest.